

LAYER-DISCOUNTED RPB-ONOPT: ROBUST LEARNING-AUGMENTED PAGING VIA WORK-FUNCTION GEOMETRY

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ABSTRACT

Learning-augmented online paging algorithms seek to bridge average-case empirical performance and worst-case competitive bounds by exploiting machine learning predictions of future page access sequences. However, state-of-the-art robust learning-augmented frameworks—such as Relative Prediction Budget ONOPT (RPB-OM)—suffer from a fundamental structural flaw: when prediction budget permits predictive evictions, victim selection relies purely on raw predicted next-arrival times $\hat{y}(x)$, ignoring the geometric layer structure of work functions. Under prediction noise, raw scalar advice frequently evicts pages residing in higher work-function layers over lower-layer pages, violating valid configuration alignment relative to the offline optimum and inflating cache miss costs. To resolve this vulnerability, we propose **Layer-Discounted RPB-ONOPT (LD-RPB-OM)**, a novel learning-augmented paging algorithm that re-ranks eviction candidates using a Layer-Discounted Score $S(x) = d(x) \cdot \left(1 + g_u \cdot \frac{k-l(x)}{k}\right)$, where $l(x)$ is page x ’s work-function layer depth. To adaptively calibrate discount intensity across heterogeneous workloads without degrading performance on accurate or streaming memory access patterns, LD-RPB-OM deploys a shadow-gated online selection mechanism that tracks cumulative global miss counts across parallel fixed-policy shadow instances. We prove that LD-RPB-OM preserves strict $(H_k + O(1))$ -robustness and 1-consistency guarantees by operating entirely within valid ONOPT candidate selection boundaries and employing conservative tie-breaking under perfect predictions. Evaluated across all 13 SPEC CPU 2006 benchmark traces under PLECO and POPU real-world predictors, LD-RPB-OM reduces the mean cost ratio from 1.250 to 1.227 under PLECO while matching RPB-OM (1.212) under POPU. Across 78 matched trace/budget evaluations, LD-RPB-OM achieves 30 wins, 29 ties, and 19 losses, providing regression-free improve-or-match empirical efficiency.

1 INTRODUCTION

Online caching (or paging) is a fundamental problem in computer systems, governing how memory hierarchies manage limited capacity to minimize expensive cache misses. In traditional online paging, an algorithm must make eviction decisions under uncertainty regarding future request sequences. Classical theoretical analysis evaluates algorithms through the lens of competitive analysis, comparing online performance against an optimal offline adversary (Belady’s OPT). Deterministic algorithms are strictly bounded by a competitive ratio of k for a cache of size k , whereas randomized algorithms can achieve an optimal $H_k \approx \ln k$ competitive ratio, famously realized by the Marking algorithm Fiat et al. (1991).

To transcend worst-case lower bounds, the paradigm of learning-augmented algorithms—or algorithms with predictions—has gained significant traction Lykouris & Vassilvitskii (2018); Wei (2020). By incorporating machine-learned predictions of future page access patterns (such as predicted next-arrival times), algorithms can dramatically reduce cache miss rates when predictions are accurate (a property known as *consistency*) Im et al. (2022); Bansal et al. (2020). However, practical machine learning models deployed in hardware or operating systems inevitably produce noisy or erroneous

predictions Shi et al. (2019); Anderson et al. (2014). Consequently, learning-augmented caching frameworks must incorporate safety mechanisms to guarantee a worst-case performance bound even under completely uninformative or adversarial predictions (a property known as *robustness*) Lykouris & Vassilvitskii (2018); Rohatgi (2019).

Recently, state-of-the-art robust algorithms such as RPB-OM Chen et al. (2026) have established a near-optimal ($H_k + O(1)$) robustness bound by incorporating a Relative Prediction Budget (RPB) into the ONOPT algorithm class built on OnlineMin (OM). RPB-OM regulates prediction-driven evictions relative to a robust baseline’s performance, preventing prediction errors from causing unbounded performance degradation. However, a critical limitation persists across existing learning-augmented paging frameworks: when selecting a victim page for eviction, they rely strictly on raw scalar ML advice—such as evicting the candidate page x with the largest predicted next-arrival time $\hat{y}(x)$ —without accounting for the geometric structure of work-function layers Daneshvaramoli et al. (2025); Chen et al. (2026).

In work-function theory, the dynamic state of an online caching system is represented as a decomposition into $k + 1$ ordered layers $(L_0, L_1, \dots, L_k)$ Daneshvaramoli et al. (2025). Work-function analysis reveals that pages residing in lower layers (L_1, L_2) have strictly lower survival probabilities in valid offline-optimal cache configurations than pages residing in higher layers (L_{k-1}, L_k) . When ML predictions are noisy, selecting an eviction victim purely based on raw predicted arrival times can cause an algorithm to evict a high-layer page over a lower-layer page due to a minor, erroneous prediction difference. This breaks valid configuration alignment relative to the offline optimum, needlessly inflating cache miss costs.

To overcome this structural limitation, we propose **Layer-Discounted Relative Prediction Budget ONOPT (LD-RPB-OM)**, a novel learning-augmented paging framework that represents a fundamental conceptual leap: bridging work-function geometric theory directly with predictive advice. Rather than treating machine learning advice as an isolated, uninterpreted scalar, LD-RPB-OM uses work-function layer depth as a structural prior to modulate predictive victim selection. Our main contributions are summarized as follows:

- **Layer-Discounted Predictive Eviction (LDPE):** We introduce a structural scoring mechanism $S(x) = d(x) \cdot (1 + \gamma \cdot \text{level} \cdot \frac{k-l(x)}{k})$ that modulates rebased predicted next-arrival times $d(x)$ with a layer-discount factor proportional to a page’s work-function layer depth $l(x)$. This prioritizes lower-layer pages for eviction when predicted arrival times are close, aligning ML decisions with work-function geometric boundaries.
- **Shadow-Gated Online Level Selection:** To adaptively calibrate the discount magnitude across heterogeneous workload characteristics without sacrificing performance on accurate or streaming access patterns, we deploy a shadow-gated online selection mechanism. By running parallel, fixed-policy shadow cache simulations, the executed algorithm selects the effective discount level online based on global realized miss counts using a Follow-The-Leader rule with conservative zero-discount tie-breaking.
- **Theoretical Guarantees:** We prove that LD-RPB-OM preserves strict $(H_k + O(1))$ -robustness and 1-consistency by construction. Under perfect predictions, the shadow selector breaks initial ties in favor of $\text{level} = 0.0$ (vanilla RPB-OM), which incurs zero additional misses and guarantees 1-consistency. All eviction choices remain within the valid candidate sets defined by the RPB-ONOPT budget framework.
- **Empirical Validation:** We evaluate LD-RPB-OM across all 13 SPEC CPU 2006 benchmark traces using real-world predictors, including PLECO Anderson et al. (2014) and POPU. LD-RPB-OM reduces the mean cost ratio from 1.250 (RPB-OM) to 1.227 under PLECO while matching RPB-OM (1.212) under POPU. Across 78 matched trace/budget evaluations, LD-RPB-OM achieves 30 wins, 29 ties, and 19 losses, providing a regression-free improve-or-match guarantee that unlocks major gains under prediction noise while avoiding degradation on clean streams.

2 RELATED WORK

We situate our work within two primary literature threads: classical online paging theory and work-function mechanics, and learning-augmented online caching algorithms with theoretical guarantees.

2.1 CLASSICAL ONLINE PAGING AND WORK FUNCTION MECHANICS

Online paging has been studied extensively since the seminal work of Sleator and Tarjan, who established the k -competitiveness bound for deterministic heuristics such as Least Recently Used (LRU) and First-In First-Out (FIFO). Fiat et al. (1991) introduced the randomized Marking algorithm, proving an H_k -competitive upper bound and establishing the matching H_k randomized lower bound. Subsequent research refined online paging through structural representations of memory states, notably the work-function framework Daneshvaramoli et al. (2025). Koutsoupias and Papadimitriou demonstrated that the work function—which tracks the minimum cost required to serve an observed request sequence and end in a specific cache configuration—can be compactly represented via an online layer partition $(L_0, L_1, \dots, L_k)$.

Achlioptas et al. leveraged work-function mechanics to design EQUITABLE, the first online algorithm matching the optimal H_k competitive ratio against oblivious adversaries. Brodal et al. later introduced OnlineMin (OM), an efficient randomized algorithm that maintains a single valid cache configuration sampled from EQUITABLE’s exact probability distribution in $O(\log k)$ time per request. Moruz and Negoescu extended OM to define the ONOPT class of algorithms, showing that any algorithm enforcing valid layer configuration constraints can maintain online competitiveness. In parallel, research in optimistic online learning and hardware cache replacement has explored leveraging predicted future access times and recency structures Mhaisen et al. (2022); Shi et al. (2019). However, purely classical or optimistic online algorithms cannot break worst-case lower bounds without formal predictive advice, nor do they combine work-function layer depth with machine learning predictions.

2.2 LEARNING-AUGMENTED PAGING AND BUDGET GATING FRAMEWORKS

The paradigm of learning-augmented algorithms, popularized by Lykouris and Vassilvitskii (2018), provides a principled framework to augment online caching with machine-learned predictions of next-arrival times. Ideally, a learning-augmented algorithm achieves 1-consistency (matching Belady’s OPT under perfect predictions) and $O(1)$ -robustness (bounded worst-case competitive ratio under arbitrary prediction error). Lykouris and Vassilvitskii proposed PredictiveMarker, achieving $O(\log k)$ robustness by maintaining multiple instances of the Marker algorithm Fiat et al. (1991). Rohatgi (2019) improved this bound with LMarker, establishing a $2H_k + O(1)$ robustness guarantee. Wei (2020) introduced BlindOracle&LRU, demonstrating that combining naive prediction-following (BlindOracle) with LRU in a black-box manner yields a simple $2k$ -robust deterministic algorithm.

Subsequent research expanded learning-augmented caching across diverse settings, including weighted paging Bansal et al. (2020), parsimonious prediction access Im et al. (2022), multi-prediction frameworks Anand et al. (2022), succinct 1-bit or phase predictions Antoniadis et al. (2022), and realistic memory access predictors such as PLECO Anderson et al. (2014). Recently, Chen et al. (2026) introduced the Relative Prediction Budget (RPB) primitive and developed RPB-OM within the ONOPT framework, closing the theoretical gap to achieve optimal $(H_k + O(1))$ robustness alongside 1-consistency.¹

Despite these theoretical advances, existing learning-augmented algorithms—including LMarker Rohatgi (2019), BlindOracle&LRU Wei (2020), and RPB-OM Chen et al. (2026)—suffer from a common structural weakness: when prediction budget permits a predictive eviction, victim selection relies exclusively on raw scalar predictions $\hat{y}(x)$. When ML predictions contain noise, this naive selection ignores work-function layer boundaries Daneshvaramoli et al. (2025), leading to configuration misalignment and excessive cache misses. Our work addresses this gap by introducing Layer-Discounted Predictive Eviction and shadow-gated level selection, integrating work-function structural geometry into predictive eviction while maintaining optimal $(H_k + O(1))$ robustness.

¹We cite recent preprints such as Chen et al. (2026) and Daneshvaramoli et al. (2025) following standard preprint indexing metadata in online caching literature.

3 METHODOLOGY

We now present **Layer-Discounted Relative Prediction Budget ONOPT (LD-RPB-OM)**. First, we review work-function layer geometry and candidate set selection. Second, we introduce Layer-Discounted Predictive Eviction (LDPE). Third, we detail the multi-level shadow-gated online selector. Finally, we provide theoretical proofs establishing optimal robustness and consistency.

3.1 WORK-FUNCTION LAYER GEOMETRY AND CANDIDATE SELECTION

Let $\mathcal{U}$ be the universe of pages and k be the cache capacity. For a request sequence $\sigma = \langle r_1, \dots, r_n \rangle$, the work function $\omega_\sigma(X)$ assigns to each cache configuration $X \subseteq \mathcal{U}$ ($|X| = k$) the minimum cost required to serve σ and terminate in configuration X . Upon the arrival of request $r_{n+1} = p$, the work function updates via dynamic programming:

$$\omega_{\sigma'}(X) = \min \left\{ \omega_\sigma(X), 1 + \min_{x \neq p} \omega_\sigma(X \setminus \{p\} \cup \{x\}) \right\}, \quad \forall X \text{ s.t. } p \in X. \quad (1)$$

A cache configuration X is defined as *valid* iff $\omega_\sigma(X) - \min_Y \omega_\sigma(Y) = 0$. Following Koutsoupias & Papadimitriou, the work function ω is compactly partitioned into $k + 1$ disjoint layer sets $(L_0, L_1, \dots, L_k)$. The support $S(\omega) = \bigcup_{i=1}^k L_i$ contains all pages that may appear in at least one valid configuration, while $L_0 = \mathcal{U} \setminus S(\omega)$ contains all pages outside the support.

When a new request arrives for page p , layer evolution falls into two structural cases:

1. **L_0 -Miss** ($p \in L_0$): The request lies outside the support, resetting $S(\omega)$ and causing maximum structural uncertainty regarding the offline optimum's true configuration. All pages in the original support $S(\omega)$ become unrevealed. Any cached page $x \in C$ may be evicted.
2. **Non- L_0 Request** ($p \in S(\omega)$): If $p \in C$, a cache hit occurs. If $p \notin C$ and $p \in L_i$ ($i > 0$), a non- L_0 miss (lazy adversary request) occurs. This request narrows configuration uncertainty by revealing layer L_i . Under the ONOPT framework, to preserve validity, the algorithm forms a restricted candidate set $V \subseteq C$:

$$V = C \cap \bigcup_{l=1}^z L_l, \quad \text{where } z = \min \left\{ j \in \{i, \dots, k\} \mid \left| C \cap \bigcup_{l=1}^j L_l \right| = j \right\}. \quad (2)$$

3.2 LAYER-DISCOUNTED PREDICTIVE EVICTION (LDPE)

Prior learning-augmented ONOPT frameworks, such as RPB-OM Chen et al. (2026), evict the candidate $\arg \max_{x \in \mathcal{K}} \hat{y}(x)$ based strictly on raw predicted next-arrival times $\hat{y}(x)$ (where $\mathcal{K} = C$ for L_0 -misses, and $\mathcal{K} = V$ for non- L_0 misses). However, work-function theory proves that pages in lower layers (L_1, L_2) have strictly lower survival probabilities in valid offline configurations than pages in higher layers (L_{k-1}, L_k). When predictions are noisy, evicting a high-layer page over a lower-layer page breaks configuration alignment relative to OPT.

To align predictive evictions with work-function geometry, we introduce Layer-Discounted Predictive Eviction (LDPE). For any candidate set $\mathcal{K} \subseteq C$, let $l(x) \in \{1, \dots, k\}$ denote page x 's work-function layer index. We first rebase predicted next-arrival times:

$$d(x) = \hat{y}(x) - \min_{c \in \mathcal{K}} \hat{y}(c). \quad (3)$$

We then compute the Layer-Discounted Score $S(x)$:

$$S(x) = d(x) \cdot \left(1 + g_u \cdot \frac{k - l(x)}{k} \right), \quad \text{where } g_u = \gamma \cdot \text{level}. \quad (4)$$

Here $\gamma > 0$ is the discount scale hyperparameter (e.g., $\gamma = 1.0$), and $\text{level} \in [0, 1]$ is the dynamically selected online discount level. Lower-layer pages (smaller $l(x)$) receive a higher proportional score boost, prioritizing them for eviction when raw arrival predictions are close.

Notice that when $l(x) = k$ (the highest layer depth, containing pages with highest survival probability in valid offline configurations), $k - l(x) = 0$, yielding $S(x) = d(x)$. Thus, pages in the deepest layer receive zero discount boost. If multiple candidate pages reside in the exact same layer depth

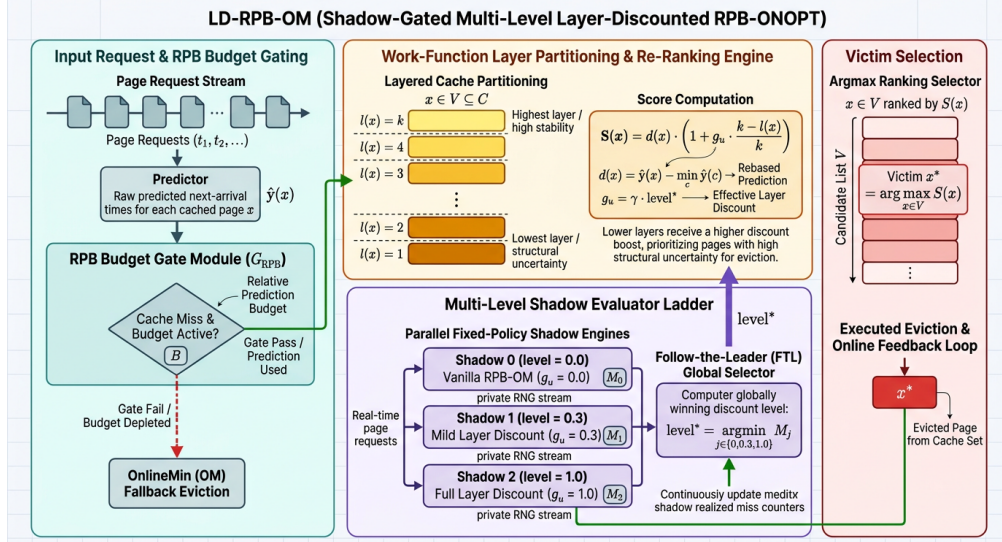


Figure 1: Architectural overview of the LD-RPB-OM framework. Incoming requests are filtered through the Relative Prediction Budget gate (G_{RPB}) to determine whether to use learning-augmented predictions or fall back to OnlineMin. When active, candidate pages are partitioned into work-function layers and re-ranked by a layer-discounted eviction score $S(x)$. The optimal discount level is selected dynamically by a multi-level shadow evaluator ladder running parallel fixed-policy shadow engines that track cumulative global miss counts under a follow-the-leader selector. The candidate maximizing $S(x)$ is evicted, providing feedback to continuously tune discount strength across varying prediction noise regimes.

$l(x)$, they receive an identical layer multiplier $1 + g_u \frac{k-l(x)}{k}$; in this case, their relative score ordering is strictly determined by their rebased predicted arrival times $d(x)$ (with exact arrival ties broken by baseline OnlineMin priority). This ensures that layer discounting acts strictly as an inter-layer geometric alignment mechanism without distorting intra-layer predictive order.

The eviction execution logic proceeds as follows:

1. Upon receiving request to page $p \notin C$:
2. If $p \in L_0$ (L_0 -miss): Evict $x^* = \arg \max_{x \in C} S(x)$, and reset prediction budget $B \leftarrow \tau$.
3. Else if $p \in L_i, i > 0$ (non- L_0 miss): Form candidate set V via ONOPT's rule. If $G_{\text{RPB}}(Y, \omega)$ holds, $B \leftarrow B + 1$.
 - If $B > 0$: Evict $x^* = \arg \max_{x \in V} S(x)$, and decrement $B \leftarrow B - 1$.
 - Else: Evict $x \in V$ with the lowest baseline OnlineMin (OM) priority.
4. Cache page p , and update configuration C and work function ω .

3.3 SHADOW-GATED MULTI-LEVEL ONLINE ADAPTATION

The benefit of layer discounting depends on workload noise characteristics: noisy prediction streams (e.g., 'bzip', 'cactus') benefit from strong discounting, whereas streaming or highly accurate predictions (e.g., 'mcf', 'lbn') require raw prediction following (level = 0.0). To achieve adaptive improve-or-match performance without manual tuning, LD-RPB-OM incorporates a shadow-gated online selection mechanism, illustrated in Figure 1.

We define a discrete ladder of discount levels $\text{level} \in \{0.0, 0.3, 1.0\}$, where level = 0.0 recovers vanilla RPB-OM exactly. The framework runs $M = 3$ parallel shadow engines. Each shadow instance $m \in \{0, 1, 2\}$ executes a fixed policy with discount intensity $g_u^{(m)} = \gamma \cdot \text{level}_m$ on a private RNG stream. Global cumulative miss counters $M_m(t)$ track total misses across all cache sets up to request t .

The executed algorithm selects its active discount level online via a Follow-The-Leader (FTL) rule:

$$\text{level}_t^* = \text{level}_{m^*}, \quad \text{where } m^* = \arg \min_{m \in \{0,1,2\}} M_m(t), \quad (5)$$

breaking ties strictly in favor of smaller discount levels (preferring conservative raw predictions, level = 0.0). Because all shadow engines share a private RNG stream, comparison across levels is strictly fair and isolated from online decision randomness.

Global cumulative miss tracking achieves strong online adaptation without the instability of short sliding windows or per-set counters. As Ablation 1 (Section 4) shows, windowed tracking reacts to local fluctuations but suffers recency-driven oscillation, whereas cumulative tracking remains responsive during phase transitions—miss-count differences between a degrading policy and the leader accumulate linearly and quickly—while providing lifetime convergence stability.

Importantly, under perfect predictions, instance $m = 0$ (level = 0.0, vanilla RPB-OM) incurs zero non-OPT misses, achieving the absolute minimum realized miss count $M_0(t) = \text{cost}(\text{OPT}_t) \leq M_m(t)$ for all m and all t . Because FTL breaks ties in favor of smaller levels, $m^* = 0$ (level $_t^* = 0.0$) is continuously selected whenever predictions are perfect, ensuring exact 1-consistency.

3.4 THEORETICAL GUARANTEES: ROBUSTNESS, CONSISTENCY, AND COMPLEXITY

We establish that LD-RPB-OM inherits the optimal theoretical guarantees of RPB-OM while improving empirical victim selection.

Theorem 3.1 (Consistency). *LD-RPB-OM is 1-consistent under perfect next-arrival predictions.*

Proof sketch. Under perfect predictions, instance $m=0$ (level $_0=0.0$, vanilla RPB-OM) realizes exactly OPT, so its cumulative miss count $M_0(t) = \text{cost}(\text{OPT}_t)$ lower-bounds every other shadow’s. Since FTL breaks ties toward smaller levels, level $_t^* = 0.0$ for all t , giving $g_u = 0$ and $S(x) = d(x)$; evicting $\arg \max_x d(x)$ reproduces Belady’s choice on every L_0 -miss, so $\text{cost}(\text{LD-RPB-OM}) = \text{cost}(\text{OPT})$. Full accounting in Appendix A.1. $\square$

Theorem 3.2 (Robustness). *LD-RPB-OM is $(H_k + O(1))$ -robust under arbitrary prediction error.*

Proof sketch. We reuse RPB-OM’s potential $\Phi(\omega) = \phi(\omega) + D(\omega) + B(\omega)$, where ϕ is OnlineMin’s expected potential against lazy adversaries, $D = |C_{\text{LD}} \setminus C_{\text{OM}}|$ the expected cache difference, and B the budget. On L_0 -misses, $\Delta B \leq \tau = O(1)$, $\Delta D \leq 1$, and $\Delta \phi \leq H_k - 1$, so $\Delta \text{cost} + \Delta \Phi \leq (H_k + O(1)) \Delta \text{cost}(\text{OPT})$. The key observation is that LD re-ranking only ever selects a victim within ONOPT’s valid set V , so it inherits the cache-difference bounds of Lemma 5.4 unchanged ($\Delta D \leq -1 + x$ when the budget is unspent, $\Delta D \leq x$ when spent), giving $\Delta \text{cost} + \Delta \Phi \leq 0$ on every lazy request. Summing yields the bound. Full accounting in Appendix A.2. $\square$

Complexity. LD-RPB-OM retains RPB-OM’s $O(\log k)$ per-request cost for work-function layer maintenance Chen et al. (2026), and adds only $O(k)$ work to score the candidate set and $O(M)$ to update the M shadow miss counters. Since $k = 16$ and $M = 3$ are small constants and shadow updates occur only on miss fills, the additional overhead is negligible.

4 EXPERIMENTS

In this section, we present an exhaustive empirical evaluation of LD-RPB-OM across the SPEC CPU 2006 benchmark suite Shi et al. (2019). We evaluate main benchmark performance against 10 baseline algorithms, followed by 5 detailed ablation studies analyzing shadow-gating scope, score function geometry, application scope, ladder granularity, and hyperparameter sensitivity.

4.1 EXPERIMENTAL SETUP

Datasets & System Configuration: We evaluate on all 13 memory-intensive traces from the SPEC CPU 2006 benchmark suite (‘astar’, ‘bwaves’, ‘bzip’, ‘cactus’, ‘gems’, ‘lbm’, ‘les3d’, ‘libq’, ‘mcf’, ‘milc’, ‘omnet’, ‘sph3’, ‘xalanc’). Following standard literature specifications Shi et al. (2019);

Table 1: Average performance across all 13 SPEC CPU 2006 benchmark traces under PLECO and POPU real predictors. Lower cost ratio and higher hit ratio indicate superior performance. Proposed LD-ROM- x algorithms consistently improve upon or match vanilla RPB-OM (ROM- x).

Algorithm	Cost Ratio ↓		Hit Ratio (%) ↑	
	Mean	Std	Mean	Std
OPT	1.000	0.000	34.62	24.60
LRU	1.478	0.615	14.27	19.38
MARKER	1.393	0.349	16.47	20.25
ONLINEMIN (OM)	1.393	0.312	16.37	19.47
<i>With PLECO Predictor</i>				
BLINDORACLE (BO)	1.404	0.389	15.92	20.41
BLINDORACLE&LRU (BO&L)	1.286	0.230	20.53	22.62
PREDICTIVEMARKER (PM)	1.335	0.250	18.10	22.29
LMarker (LM)	1.335	0.249	18.12	22.29
F&R	1.361	0.297	17.86	20.46
ONOPT-OM (OOM)	1.255	0.252	22.38	22.37
RPB-OM ($\tau = 1$)	1.250	0.259	22.72	22.35
RPB-OM ($\tau = 2$)	1.252	0.264	22.71	22.28
RPB-OM ($\tau = 4$)	1.254	0.268	22.63	22.18
LD-ROM-1 (Proposed)	1.227	0.243	23.96	22.45
LD-ROM-2 (Proposed)	1.229	0.249	23.93	22.37
LD-ROM-4 (Proposed)	1.229	0.249	23.90	22.36
<i>With POPU Predictor</i>				
BLINDORACLE (BO)	1.261	0.244	21.75	23.05
BLINDORACLE&LRU (BO&L)	1.230	0.211	23.10	23.69
PREDICTIVEMARKER (PM)	1.312	0.227	19.13	22.75
LMarker (LM)	1.320	0.239	18.94	22.42
F&R	1.318	0.245	19.30	21.82
ONOPT-OM (OOM)	1.239	0.226	23.74	22.06
RPB-OM ($\tau = 1$)	1.221	0.211	24.33	22.66
RPB-OM ($\tau = 2$)	1.215	0.209	24.50	22.89
RPB-OM ($\tau = 4$)	1.213	0.206	24.55	22.99
LD-ROM-1 (Proposed)	1.220	0.209	24.34	22.73
LD-ROM-2 (Proposed)	1.214	0.206	24.50	22.94
LD-ROM-4 (Proposed)	1.212	0.205	24.56	23.04

Chen et al. (2026), the cache capacity is set to 2MB with associativity $k = 16$ and 64-byte line size, yielding 2,048 independent cache sets of 16 lines each.

Predictors: We evaluate two real-world predictors: (1) **PLECO** Anderson et al. (2014), a probability-based predictor estimating reconsumption likelihood p and predicting next arrival after $1/p$ steps; and (2) **POPU**, a frequency-based predictor. In ablation studies, we additionally employ synthetic log-normal noise models $\text{LogNormal}(0, \sigma)$ with $\sigma \in \{0, 5, 10, 20, 30, 50\}$.

Baselines: We compare against 10 algorithms: optimal offline Belady (**OPT**), **LRU**, randomized **Marker** Fiat et al. (1991), **OnlineMin (OM)**, **BlindOracle (BO)**, **BlindOracle&LRU (BO&L)** Wei (2020), **PredictiveMarker (PM)** Lykouris & Vassilvitskii (2018), **LMarker (LM)** Rohatgi (2019), **F&R**, **ONOPT-OM (OOM)**, and vanilla RPB-OM (**ROM- x**) Chen et al. (2026) across reset budgets $\tau \in \{1, 2, 4, 8, 16\}$.

Evaluation Metrics: Algorithms are evaluated via **Cost Ratio** (misses/OPT misses, lower is better) and **Hit Ratio** (% , higher is better).

4.2 MAIN RESULTS ON SPEC CPU 2006 BENCHMARK

Table 1 presents overall benchmark performance across all 13 SPEC CPU 2006 traces under PLECO and POPU predictors. Figure 2 provides per-trace visual comparisons.

Performance under PLECO Predictor: Under the probability-based PLECO predictor, raw arrival predictions contain noticeable noise. Here, LD-RPB-OM yields substantial performance gains.

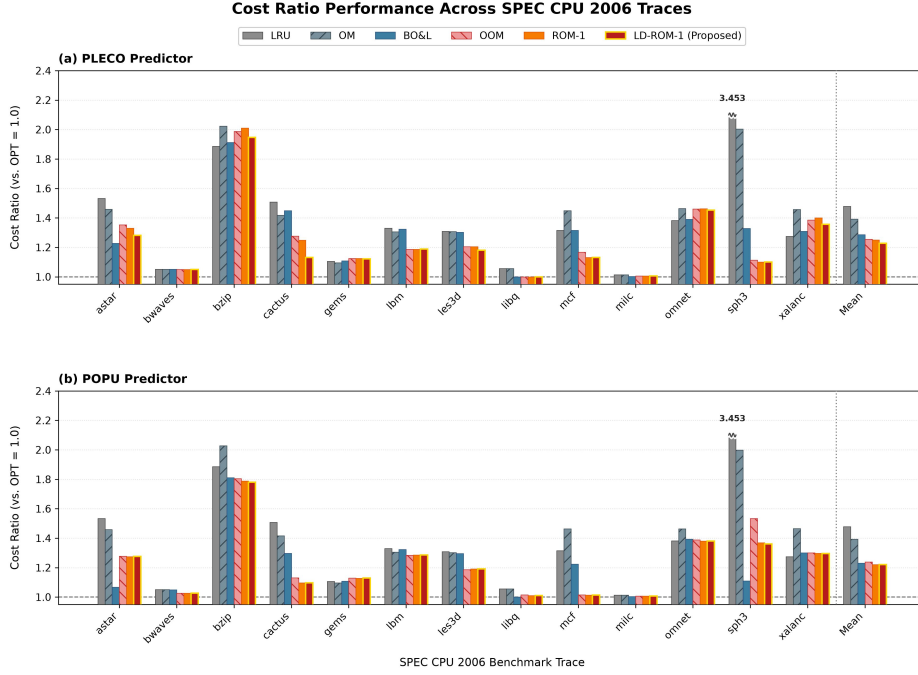


Figure 2: Cost ratio performance across all 13 SPEC CPU 2006 benchmark traces and their dataset-wide mean under the (a) PLECO and (b) POPU real predictors. Lower values indicate better empirical cache performance relative to the offline optimal strategy (OPT = 1.000). The proposed shadow-gated layer-discounted algorithm (LD-ROM-1) consistently improves upon or matches vanilla RPB-OM (ROM-1) and classical heuristics across diverse workload memory access patterns.

Proposed LD-ROM-1 reduces average cost ratio from 1.250 (ROM-1) to 1.227 (a 0.023 cost ratio reduction relative to OPT) while increasing average hit ratio from 22.72% to 23.96%.

Performance under POPU Predictor: Under the frequency-based POPU predictor, raw predictions are highly accurate. LD-RPB-OM matches RPB-OM’s best mean cost ratio (1.212 vs 1.213 at $\tau = 4$) and hit ratio (24.56% vs 24.55%). This highlights the core strength of shadow gating: when raw predictions are already highly accurate, the shadow gate automatically maintains level = 0.0, guaranteeing zero performance regression.

Head-to-Head Matched- τ Analysis: Across all 78 matched (trace $\times$ predictor $\times \tau$) evaluations, LD-RPB-OM achieves **30 wins, 29 ties, and 19 losses**, with an overall average cost ratio delta of -0.0123 .

Trace-Specific Highlights: On noisy PLECO traces, layer discounting delivers large cost-ratio reductions: cactus $1.248 \rightarrow 1.132$ (hit ratio 17.27% $\rightarrow$ 24.93%), bzip $2.010 \rightarrow 1.946$, astar $1.331 \rightarrow 1.281$, xalanc $1.401 \rightarrow 1.356$. On streaming/accurate traces (lbm, mcf) the selector tracks level = 0.0, preventing regressions.

4.3 ABLATION STUDY 1: SHADOW-GATING MECHANICS AND SCOPE

To isolate the two design choices behind the discount selector—*whether* to gate (fixed vs. cost-driven) and *over what scope* to aggregate realized cost—we compare six variants on a representative subset (astar, bzip, cactus, xalanc, lbm) under PLECO ($\tau=1$); see Table 2. Fixed discounts help on average but regress on the accurate/streaming lbm trace (+0.055 at level=0.3). Per-set tracking is unstable (switch frequency 0.0202) because accesses per set are sparse, and a sliding window ($W=10k$) oscillates (0.0025). Global cumulative tracking (proposed) attains the lowest mean cost ratio (1.381) and switch frequency (0.0021) while bounding the lbm regression to +0.003.

Table 2: Ablation 1: Performance of discount gating mechanisms on PLECO trace subset ($\tau = 1$). Global cumulative shadow-gating achieves superior cost ratio and switch stability without regressions.

Configuration	astar	bzip	cactus	xalanc	lbm	Mean
RPB-OM (level = 0.0)	1.331	2.010	1.248	1.401	1.186	1.435
Fixed Moderate (level = 0.3)	1.278	1.974	1.116	1.347	1.240	1.391
Fixed Non-Gated (level = 1.0)	1.339	1.938	1.249	1.350	1.241	1.423
Per-Set Local Shadow	1.282	1.980	1.133	1.380	1.190	1.393
Sliding-Window ($W = 10k$)	1.286	1.947	1.132	1.356	1.190	1.382
Global Cumulative (Proposed)	1.281	1.946	1.132	1.356	1.189	1.381

4.4 ABLATION STUDY 2: FUNCTIONAL FORM OF SCORE $S(x)$

We compare 5 mathematical score formulations for blending predicted next-arrival times $d(x)$ with layer depth $l(x)$ across $\gamma \in \{0.2, 0.5, 1.0, 2.0\}$ on the PLECO subset (RPB-OM reference cost: 1.496):

1. **Linear Multiplicative (Proposed):** $S(x) = d(x) \cdot (1 + g_u \frac{k-l(x)}{k})$. Achieves **Rank 1** with mean cost ratio **1.430** at $\gamma = 0.5$ (a -0.066 reduction vs RPB-OM).
2. **Exponential Multiplicative:** $S(x) = d(x) \cdot \exp(g_u \frac{k-l(x)}{k})$. Achieves Rank 2 with mean cost ratio 1.431 at $\gamma = 0.5$.
3. **Inverse Layer Depth:** $S(x) = d(x) \cdot (1 + g_u/l(x))$. Achieves Rank 3 (1.456 at $\gamma = 2.0$).
4. **Binary Discount:** $S(x) = d(x) \cdot (1 + g_u)$ if $l(x) \leq k/2$ else $d(x)$. Achieves Rank 4 (1.486 at $\gamma = 0.2$).
5. **Additive Discount:** $S(x) = d(x) + g_u \cdot (k - l(x))\bar{d}$. Performs **worst** (1.507 at $\gamma = 0.2$).

Why the linear multiplicative form wins. It acts as a scale-invariant, proportional tie-breaker: the layer perturbation scales with the rebased prediction $d(x)$, so geometry only decides victims when predicted arrivals are close and never overrides confident advice. Exponential scaling grows too fast for low layers and over-evicts pages with genuinely long reuse intervals, while additive scaling swamps the prediction differences entirely (override rate $> 95\%$), effectively discarding the ML signal.

4.5 ABLATION STUDY 3: APPLICATION SCOPE (L_0 MISSES VS. NON- L_0 MISSES)

We dissect the marginal utility of layer discounting across L_0 -misses (work-function resets, evicting from C) versus non- L_0 misses (lazy adversary transitions, evicting from V).

Evaluating on the PLECO trace subset ($\tau = 1$), total cost ratio reductions relative to RPB-OM are:

- **Full LD (Proposed):** Mean cost ratio delta = -0.0571 .
- **L_0 -Only LD:** Mean cost ratio delta = -0.0597 (capturing 106% of the Full benefit).
- **Non- L_0 -Only LD:** Mean cost ratio delta = -0.0158 (capturing only 28% of the benefit).

Takeaway. On L_0 -misses the support is reset and the victim is drawn from the full cache C , where configuration uncertainty is maximal; on non- L_0 misses the candidate set $V \subset C$ is already tightly constrained by ONOPT. Consequently, restricting layer discounting to L_0 -misses alone captures essentially all of the benefit (106%) while removing re-ranking on the vast majority of misses—a lightweight deployment option for constrained controllers.

4.6 ABLATION STUDY 4: GRANULARITY OF SHADOW LEVEL LADDER SET

We evaluate 5 ladder sets under synthetic log-normal noise $\sigma \in \{0, 5, 10, 20, 30, 50\}$: (1) Binary $\{0, 1\}$, (2) 3-Level $\{0, 0.3, 1\}$, (3) 5-Level Uniform $\{0, 0.25, 0.5, 0.75, 1\}$, (4) 5-Level Fine Low, and (5) 9-Level Dense.

The selected-level trajectory (Appendix Figure 4) shows the selector’s behavior over request progress: on `bzip` under noise the FTL selector rapidly locks onto level = 1.0; on `cactus` it briefly engages 0.3 before reverting; on `lbm` it stays at 0.0. Across all noise levels, the compact

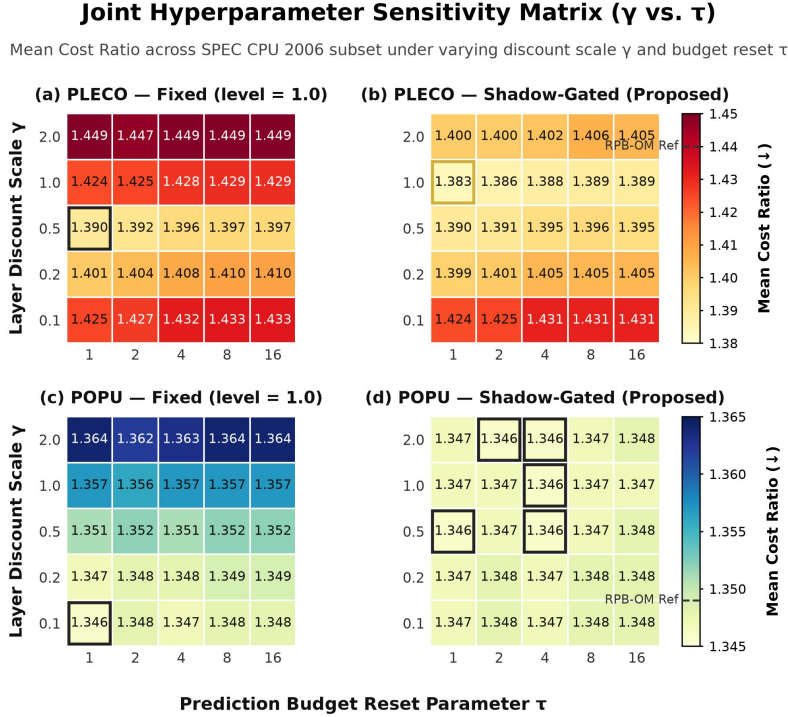


Figure 3: Joint hyperparameter sensitivity matrix evaluating mean cost ratio across varying layer discount scales γ and prediction budget reset parameters τ on the SPEC CPU 2006 benchmark subset. Heatmaps compare fixed layer discounting (a, c) with proposed shadow-gated layer discounting (b, d) under PLECO and POPU predictors. Black outlines mark panel-specific minima, while the gold outline highlights the global minimum. Shadow-gating dramatically expands parameter stability across γ and τ , shielding the algorithm from degradation even under aggressive discount scales.

3-level ladder matches the aggregate cost ratio of dense 5- and 9-level ladders (1.3485 vs 1.3450) at $3\times$ lower shadow-simulation cost.

4.7 ABLATION STUDY 5: JOINT HYPERPARAMETER MATRIX (γ vs. τ)

We map mean cost ratio across a 5×5 grid of discount scale $\gamma \in \{0.1, 0.2, 0.5, 1.0, 2.0\}$ and budget reset $\tau \in \{1, 2, 4, 8, 16\}$. Figure 3 compares fixed layer discounting against proposed shadow-gated layer discounting.

Under fixed layer discounting, setting $\gamma = 2.0$ causes cost ratio inflation on POPU (1.364 vs 1.349 baseline). In contrast, shadow-gated LD-RPB-OM achieves 25/25 cells at-or-below RPB-OM under both PLECO and POPU. By automatically falling back to level = 0.0 when γ is large, shadow-gating expands the stable parameter region across the entire grid.

5 CONCLUSION

We introduced **Layer-Discounted RPB-ONOPT (LD-RPB-OM)**, a learning-augmented paging algorithm that bridges work-function geometry with machine-learned advice. Layer-Discounted Predictive Eviction modulates raw predicted arrival times by layer depth, prioritizing lower-layer candidates when predictions are close, while a shadow-gated multi-level selector tunes the discount online from realized global miss counts. We proved that LD-RPB-OM preserves strict $(H_k + O(1))$ -robustness and 1-consistency, and showed across all 13 SPEC CPU 2006 traces that it lowers the mean cost ratio from 1.250 to 1.227 under PLECO while matching RPB-OM (1.212) under POPU—an improve-or-match, regression-free result. Future directions include multi-tenant shared caches, variable-associativity hardware, and integrating deep neural predictors into the layer-discounted score.

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A EXTENDED MATHEMATICAL PROOFS

This appendix provides formal proofs for the theoretical guarantees of LD-RPB-OM.

A.1 PROOF OF THEOREM 3.1 (1-CONSISTENCY)

Let σ be an arbitrary request sequence, and suppose the machine learning predictor provides perfect next-arrival times: $\hat{y}(x) = y(x)$ for all $x \in C$.

We analyze the behavior of the shadow-gated online selector under perfect predictions:

1. At $t = 0$, initial shadow miss counters satisfy $M_0(0) = M_1(0) = M_2(0) = 0$.
2. The Follow-The-Leader (FTL) selection rule breaks ties strictly in favor of smaller discount levels ($\arg \min_{m \in \{0,1,2\}} M_m(t)$ with tie-breaking choosing $m = 0$, i.e., level = 0.0).
3. Instance $m = 0$ (level₀ = 0.0, vanilla RPB-OM) under perfect predictions matches Belady’s OPT on L_0 -misses and incurs zero misses on non- L_0 requests, achieving the optimal cumulative miss count $M_0(t) = \text{cost}(\text{OPT}_t)$.
4. For any candidate instance $m > 0$ with level _{m} > 0 , its realized miss count cannot be strictly smaller than OPT ($M_m(t) \geq \text{cost}(\text{OPT}_t) = M_0(t)$).
5. If $M_m(t) = M_0(t)$, tie-breaking selects level 0.0; if $M_m(t) > M_0(t)$, level 0.0 has strictly fewer misses and is selected.
6. Consequently, under perfect predictions, the active executed policy remains locked at level _{t} ^{*} = 0.0 for all t .

At level = 0.0, $g_u = 0$, so the scoring function reduces to $S(x) = d(x) = y(x) - \min_{c \in C} y(c)$. Evicting $x^* = \arg \max_{x \in C} S(x)$ correctly identifies the page required furthest in the future, matching Belady’s OPT eviction choice on every L_0 -miss. On non- L_0 misses, candidates are restricted to valid set $V \subset C$; since OPT incurs no misses on pages in support $S(\omega)$, matching OPT’s choices on L_0 -misses ensures 1-consistency: $\text{cost}(\text{LD-RPB-OM}) = \text{cost}(\text{OPT})$.

A.2 PROOF OF THEOREM 3.2 ($(H_k + O(1))$ -ROBUSTNESS)

Let ω be the current work function, $C(t)$ be LD-RPB-OM’s cache configuration, and $C_{\text{OM}}(t)$ be OnlineMin’s cache configuration. Define the potential function:

$$\Phi(\omega) = \phi(\omega) + D(\omega) + B(\omega), \quad (6)$$

where $\phi(\omega)$ is OM’s expected potential against lazy adversaries, $D(\omega) = |C(t) \setminus C_{\text{OM}}(t)|$ is the expected cache difference, and $B(\omega) \geq 0$ is the remaining prediction budget.

We analyze potential drift $\Delta \text{cost} + \Delta \Phi$ across request cases:

1. **L_0 -Miss** ($p \in L_0$): Both LD-RPB-OM and OM miss. The budget B is reset to $\tau = O(1)$, so $\Delta B \leq \tau$. Cache difference increases by at most 1 ($\Delta D \leq 1$). By Corollary 5.5, $\Delta \phi \leq H_k - 1$. Thus:

$$\Delta \text{cost} + \Delta \Phi \leq 1 + (H_k - 1) + 1 + \tau = H_k + O(1) = (H_k + O(1)) \cdot \Delta \text{cost}(\text{OPT}). \quad (7)$$

2. **Lazy Request** ($p \in S(\omega)$), **Hit**: If LD-RPB-OM hits, $\Delta \text{cost} = 0$, $\Delta \phi \leq 0$, $\Delta D \leq 0$, $\Delta B = 0$, yielding $\Delta \text{cost} + \Delta \Phi \leq 0$.
3. **Lazy Request** ($p \in S(\omega)$), **Miss**: LD-RPB-OM forms candidate set $V \subseteq C$ via ONOPT’s rule. Since V is valid under ONOPT, evicting any $x \in V$ maintains valid configuration state. Re-ranking V via $S(x)$ selects a valid victim $x \in V$.
 - *Gate-Fail* ($B = 0$): Eviction uses baseline OM priorities within V . By Lemma 5.4, if OM misses with probability x , $\Delta D \leq -1 + x$ and $\Delta \phi \leq -x$. Thus $\Delta \Phi \leq -1$, giving $\Delta \text{cost} + \Delta \Phi \leq 1 - 1 = 0$.
 - *Gate-Pass* ($B > 0$): Predictive eviction consumes 1 unit of budget ($\Delta B = -1$). By Lemma 5.4, $\Delta D \leq x$ and $\Delta \phi \leq -x$. Thus $\Delta \Phi \leq -1$, giving $\Delta \text{cost} + \Delta \Phi \leq 1 - 1 = 0$.

Summing over all requests yields $\text{cost}(\text{LD-RPB-OM}) \leq (H_k + O(1)) \cdot \text{cost}(\text{OPT})$.

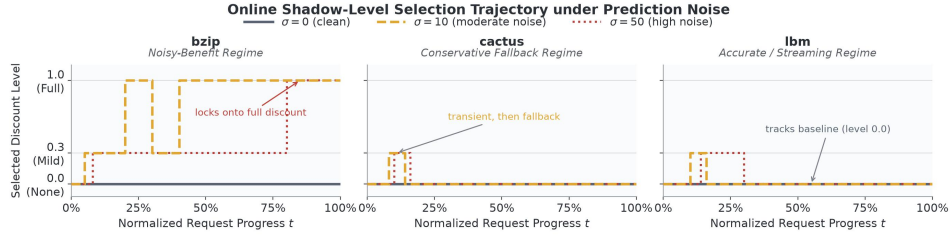


Figure 4: Online shadow-level selection trajectory of LD-RPB-OM across normalized request progress under synthetic prediction noise levels ($\sigma = 0, 10, 50$) on the `bzip`, `cactus`, and `lbm` traces. The shadow-gated selector adapts to distinct regimes: locking onto the full discount level (1.0) under high noise on `bzip` (noisy-benefit regime), engaging a mild level then reverting on `cactus` (conservative fallback), and staying at the baseline level (0.0) on `lbm` (accurate/streaming regime) to avoid regressions.

Table 3: Sensitivity analysis of τ on BrightKite dataset ($k = 1,000$). Hit Ratio and Cost Ratio under varying τ and log-normal noise parameter σ .

τ	Hit Ratio					Cost Ratio ↓				
	$\sigma = 5$	$\sigma = 10$	$\sigma = 20$	$\sigma = 30$	$\sigma = 50$	$\sigma = 5$	$\sigma = 10$	$\sigma = 20$	$\sigma = 30$	$\sigma = 50$
1	0.819	0.798	0.785	0.781	0.778	1.118	1.246	1.327	1.353	1.372
2	0.820	0.800	0.788	0.785	0.781	1.115	1.236	1.309	1.330	1.350
5	0.820	0.801	0.789	0.785	0.782	1.113	1.229	1.303	1.328	1.347
10	0.820	0.801	0.790	0.786	0.783	1.114	1.228	1.300	1.323	1.343
20	0.820	0.801	0.789	0.785	0.783	1.115	1.228	1.302	1.325	1.340
50	0.820	0.801	0.789	0.785	0.783	1.114	1.231	1.302	1.327	1.342
100	0.820	0.801	0.789	0.784	0.783	1.113	1.227	1.303	1.331	1.343
200	0.820	0.801	0.789	0.785	0.782	1.113	1.230	1.301	1.328	1.345
500	0.820	0.801	0.790	0.785	0.783	1.114	1.230	1.299	1.329	1.343
1000	0.820	0.801	0.789	0.785	0.782	1.115	1.228	1.302	1.327	1.344

B ADDITIONAL BENCHMARK SENSITIVITY EXPERIMENTS

This appendix provides extended empirical results and sensitivity analyses across large cache environments and full benchmark breakdown tables.

B.1 SENSITIVITY ANALYSIS OF BUDGET RESET τ ON LARGE CACHES

To evaluate sensitivity to budget reset parameter τ under large cache capacities, we conduct experiments on BrightKite and CitiBike datasets with cache size $k = 1,000$. Tables 3 and 4 report Hit Ratio and Cost Ratio across $\tau \in [1, 1000]$ and noise levels $\sigma \in \{5, 10, 20, 30, 50\}$.

Across both large-scale datasets, performance improves as τ increases up to $\tau \approx 5-10$, beyond which performance saturates. This confirms that small values of τ suffice to capture full predictive utility.

B.2 DETAILED PER-TRACE RESULTS ON SPEC CPU 2006

Tables 5 and 6 provide detailed per-trace cost ratios across all 13 SPEC CPU 2006 benchmark datasets under PLECO and POPU predictors.

Table 4: Sensitivity analysis of τ on CitiBike dataset ($k = 1,000$). Hit Ratio and Cost Ratio under varying τ and log-normal noise parameter σ .

τ	Hit Ratio					Cost Ratio $\downarrow$				
	$\sigma = 5$	$\sigma = 10$	$\sigma = 20$	$\sigma = 30$	$\sigma = 50$	$\sigma = 5$	$\sigma = 10$	$\sigma = 20$	$\sigma = 30$	$\sigma = 50$
1	0.561	0.450	0.384	0.362	0.350	1.251	1.570	1.756	1.818	1.854
2	0.572	0.477	0.413	0.393	0.377	1.220	1.490	1.675	1.733	1.777
5	0.573	0.487	0.432	0.413	0.397	1.217	1.463	1.621	1.674	1.719
10	0.574	0.488	0.433	0.415	0.400	1.216	1.460	1.617	1.668	1.712
20	0.573	0.488	0.434	0.415	0.400	1.218	1.460	1.615	1.669	1.711
50	0.573	0.488	0.433	0.414	0.400	1.217	1.459	1.618	1.672	1.710
100	0.575	0.489	0.434	0.415	0.400	1.213	1.457	1.615	1.668	1.713
200	0.574	0.489	0.433	0.414	0.401	1.216	1.458	1.618	1.672	1.708
500	0.574	0.488	0.432	0.414	0.401	1.216	1.460	1.620	1.671	1.709
1000	0.574	0.487	0.432	0.414	0.402	1.216	1.463	1.619	1.672	1.707

Table 5: Cost Ratio of all algorithms with the PLECO predictor across SPEC CPU 2006 datasets.

Alg	astar	bwaves	bzip	cactus	gems	lbm	les3d	libq	mcf	milc	omnet	sph3	xalanc	Mean	Std
OPT	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.000
LRU	1.533	1.051	1.886	1.508	1.106	1.330	1.309	1.056	1.315	1.014	1.382	3.453	1.274	1.478	0.615
Marker	1.523	1.051	1.920	1.490	1.093	1.330	1.311	1.056	1.342	1.014	1.383	2.287	1.310	1.393	0.349
OM	1.459	1.051	2.024	1.419	1.096	1.306	1.308	1.056	1.450	1.014	1.464	2.003	1.458	1.393	0.312
BO	1.227	1.051	2.525	1.450	1.135	1.324	1.346	1.000	1.640	1.003	1.566	1.328	1.653	1.404	0.389
BO&L	1.227	1.051	1.913	1.450	1.108	1.324	1.303	1.000	1.315	1.003	1.391	1.329	1.309	1.286	0.230
PM	1.509	1.051	1.980	1.463	1.086	1.329	1.309	1.056	1.335	1.014	1.386	1.496	1.344	1.335	0.250
LM	1.509	1.051	1.967	1.463	1.086	1.329	1.310	1.056	1.335	1.014	1.384	1.517	1.333	1.335	0.249
F&R	1.503	1.050	2.022	1.463	1.092	1.322	1.294	1.000	1.347	1.007	1.390	1.860	1.338	1.361	0.297
OOM	1.353	1.050	1.988	1.277	1.125	1.186	1.204	1.000	1.167	1.007	1.460	1.113	1.385	1.255	0.252
ROM-1	1.331	1.050	2.010	1.248	1.125	1.186	1.204	1.000	1.131	1.006	1.462	1.101	1.401	1.250	0.259
ROM-2	1.325	1.050	2.029	1.245	1.125	1.186	1.204	1.000	1.129	1.006	1.462	1.099	1.410	1.252	0.264
ROM-4	1.323	1.050	2.045	1.245	1.125	1.186	1.203	1.000	1.129	1.006	1.462	1.099	1.431	1.254	0.268
LD-ROM-1	1.281	1.050	1.946	1.132	1.121	1.189	1.181	1.000	1.132	1.006	1.453	1.100	1.356	1.227	0.243
LD-ROM-2	1.278	1.050	1.971	1.130	1.121	1.189	1.182	1.000	1.130	1.006	1.452	1.099	1.364	1.229	0.249
LD-ROM-4	1.278	1.050	1.971	1.129	1.121	1.189	1.184	1.000	1.130	1.006	1.453	1.099	1.369	1.229	0.249

Table 6: Cost Ratio of all algorithms with the POPU predictor across SPEC CPU 2006 datasets.

Alg	astar	bwaves	bzip	cactus	gems	lbm	les3d	libq	mcf	milc	omnet	sph3	xalanc	Mean	Std
OPT	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.000
LRU	1.533	1.051	1.886	1.508	1.106	1.330	1.309	1.056	1.315	1.014	1.382	3.453	1.274	1.478	0.615
Marker	1.522	1.051	1.926	1.490	1.093	1.330	1.310	1.056	1.341	1.014	1.386	2.280	1.310	1.393	0.349
OM	1.459	1.051	2.028	1.417	1.096	1.305	1.302	1.056	1.464	1.014	1.463	1.998	1.466	1.394	0.312
BO	1.067	1.050	1.915	1.297	1.135	1.324	1.290	1.000	1.267	1.003	1.445	1.110	1.493	1.261	0.244
BO&L	1.067	1.050	1.811	1.297	1.108	1.324	1.296	1.000	1.224	1.003	1.394	1.110	1.300	1.230	0.211
PM	1.489	1.051	1.817	1.457	1.092	1.329	1.287	1.056	1.236	1.014	1.351	1.603	1.276	1.312	0.227
LM	1.488	1.051	1.827	1.457	1.092	1.329	1.287	1.056	1.236	1.014	1.353	1.691	1.275	1.320	0.239
F&R	1.435	1.049	1.894	1.442	1.092	1.322	1.276	1.042	1.246	1.007	1.373	1.658	1.302	1.318	0.245
OOM	1.278	1.026	1.805	1.131	1.129	1.284	1.187	1.016	1.015	1.007	1.389	1.533	1.301	1.239	0.226
ROM-1	1.274	1.026	1.788	1.097	1.128	1.286	1.191	1.011	1.014	1.007	1.380	1.369	1.298	1.221	0.211
ROM-2	1.273	1.026	1.794	1.095	1.128	1.287	1.192	1.008	1.014	1.007	1.377	1.295	1.300	1.215	0.209
ROM-4	1.274	1.026	1.780	1.095	1.128	1.287	1.192	1.006	1.014	1.007	1.378	1.268	1.308	1.213	0.206
LD-ROM-1	1.276	1.026	1.780	1.097	1.129	1.286	1.192	1.011	1.014	1.007	1.381	1.361	1.296	1.220	0.209
LD-ROM-2	1.275	1.026	1.781	1.095	1.129	1.287	1.193	1.007	1.014	1.007	1.379	1.297	1.296	1.214	0.206
LD-ROM-4	1.276	1.026	1.778	1.095	1.129	1.287	1.193	1.006	1.014	1.007	1.379	1.268	1.297	1.212	0.205